

Objective:

Develop an AI-based system that monitors network traffic, predicts congestion, and recommends bandwidth reallocation to optimize network performance.

1. Input Data Details

Dataset Fields:

- **Timestamp:** Date and time of the log entry.
- **Device Name:** Identifier for the router (e.g., Router_A, Router_B).
- **Source IP:** IP address initiating traffic.
- **Destination IP:** IP address receiving traffic.
- **Traffic Volume (MB/s):** Amount of data transferred per second.
- **Latency (ms):** Delay experienced in the network.
- **Bandwidth Allocated (MB/s):** Total bandwidth assigned to the device.
- **Bandwidth Used (MB/s):** Actual bandwidth consumption.
- **Congestion Flag:** Indicator if congestion, high utilization, or outage occurred ("Yes"/"No").
- **Log Text:** Descriptive message about the event.

Preparation Tips:

- Use the dataset to extract features such as traffic volume, latency, bandwidth usage, and time-based patterns.
 - Label data points with congestion events for supervised learning.
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2. Expected Outputs

Primary Goals:

- **Prediction Model:**
 - **Input:** Recent network metrics (e.g., last N hours of traffic volume, latency, bandwidth used).
 - **Output:** Probability or binary prediction of congestion or high utilization in the next time interval.
- **Reallocation Recommendations:**

- Based on predictions, suggest bandwidth adjustments (e.g., increase/decrease bandwidth for specific devices or links).

Additional Outputs:

- Visual dashboards showing real-time network status.
 - Alerts for impending congestion or outages.
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3. Key Requirements & Guidelines

- **Data Handling:**
 - Use historical data to train models.
 - Validate models with a subset of data.
 - Test predictions on unseen data points.
- **Modeling Approaches:**
 - Time-series forecasting (e.g., ARIMA, LSTM).
 - Classification models (e.g., Random Forest, Logistic Regression) for congestion prediction.
 - Anomaly detection techniques for outages.
- **Features to Consider:**
 - Traffic volume trends.
 - Latency patterns.
 - Bandwidth utilization ratios.
 - Time-of-day or day-of-week patterns.
- **Evaluation Metrics:**
 - Accuracy, Precision, Recall for classification.
 - RMSE or MAE for forecasting.
 - Confusion matrix for event prediction.
- **Deliverables:**
 - Trained model(s) with code/scripts.
 - A brief report explaining the approach, features used, and results.
 - A demo or visualization of predictions and recommendations.

4. Additional Tips for Teams

- Focus on creating a scalable and interpretable model.
- Use visualization tools (e.g., matplotlib, seaborn) to illustrate network patterns.
- Consider real-time prediction capabilities for practical deployment.
- Document assumptions and limitations.